

homework-03

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Part 1. Homework Set Up

```
# loading in packages
library(tidyverse) # loading in tidyverse
library(janitor) # loading in janitor
library(readxl) # loading in readxl
library(ggeffects) # loading in ggeffects

# reading in and storing data as object "kelp"
kelp <- read_csv("data/temp-kelp.csv") # data/ in order to direct the file path
# reading in and storing personal data as object "personal"
my_data <- read_csv("data/personal-data.csv") # data/ in order to direct the file path
```

Part 2. Problems

Problem 1. Giant kelp fronds

a. An appropriate test

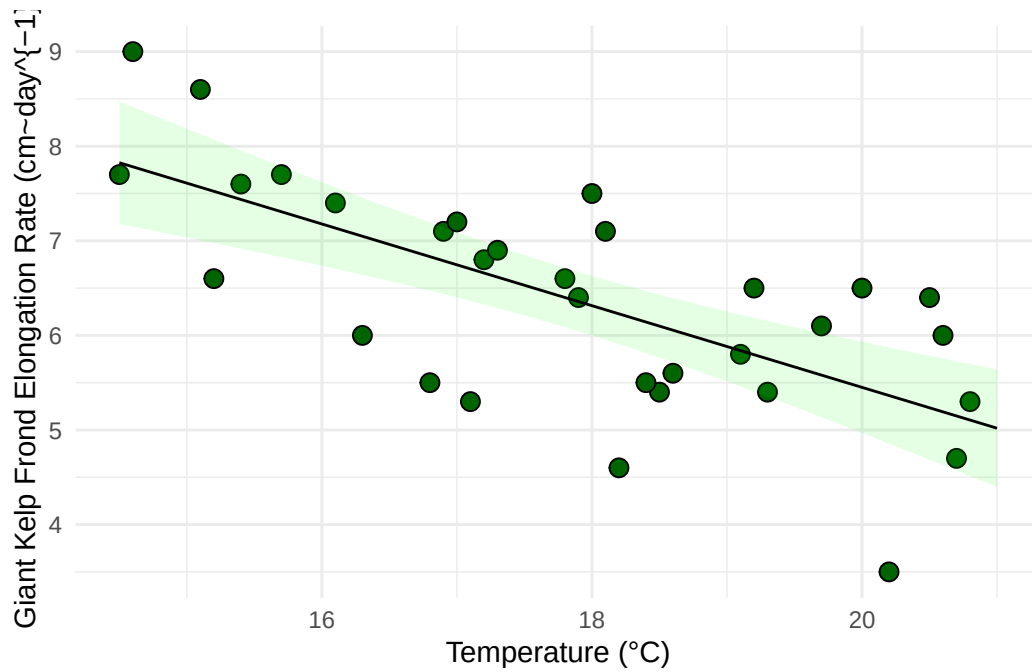
In our study, we are trying to measure the strength of the relationship between two variables (ocean temperature and giant kelp frond elongation rate). In order to test the strength of this relationship, we should use a Pearson's product-moment correlation and a Spearman's rank correlation. Pearson's correlation test is the parametric version of the test, while Spearman's rank correlation is the non-parametric version that measures the correlation between ranks (rather than continuous variables).

b. Create a visualization

```
# fitting the linear model to generate the ribbon + linear line
kelp_model <- lm(
  kelp_elong ~ temp_c, # formula: kelp growth rate as a function of temp
  data = kelp) # data from the kelp data set

# generating predictor values for the linear model + ribbon
kelp_preds <- ggpredict(
  kelp_model,          # model object
  terms = "temp_c")   # predictor (in quotation marks)

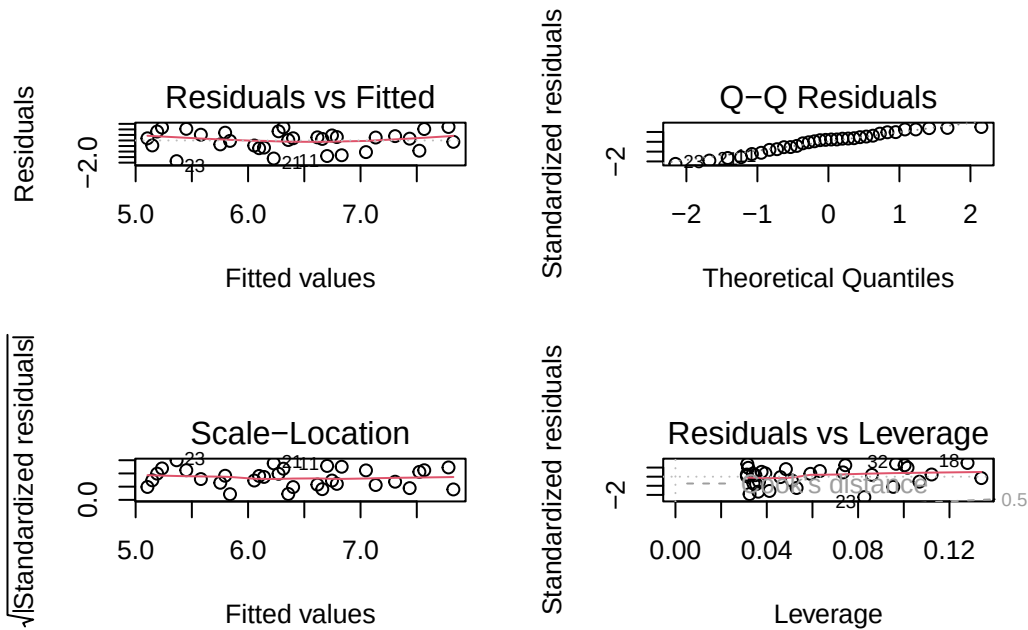
# generating the visualization
# base layer: ggplot
ggplot(data = kelp, # data from kelp data set
       mapping = aes(x = temp_c, # x-axis = temp
                     y = kelp_elong)) + # y-axis = growth rate
# first layer: geom_point
geom_point(size = 3, # changing point size
          fill = "darkgreen", # adding data points in the color green
          shape = 21) + # point shape = circles
# second layer: ribbon representing confidence interval
# using predictions data frame
geom_ribbon(data = kelp_preds, # data from predicted data set
          aes(x = x, # x = x in the data set
              y = predicted, # y = y in the data set
              ymin = conf.low, # min y point on ribbon set by CI
              ymax = conf.high), # max y point on ribbon set by CI
          alpha = 0.1, # opaqueness of the ribbon
          fill = "green") + # changing ribbon color
# third layer: line representing model predictions
# using predictions data frame
geom_line(data = kelp_preds, # data from predicted data set
         aes(x = x, # x = x in the data set
             y = predicted)) + # y = y in the data set
# labeling x- and y-axes
labs(x = "Temperature (°C)", # x = "Temperature (°C)"
     y = "Giant Kelp Frond Elongation Rate (cm~day^{-1})") + # y = elongation rate
# changing theme to be different than the default
theme_minimal()
```



c. Check your assumptions and run your test

Checking Assumptions

```
# base r residuals plots required to check the assumptions
par(mfrow = c(2,2))
plot(kelp_model)
```



My tests check the assumptions of fitting a linear model using the Residuals Vs Fitted, Q-Q Residuals, Scale-Location and Residuals vs. Leverage plots to test the homoscedasticity of residuals, the normality of the residuals, and the influence of outliers on the model. Based on the model, the residuals roughly fit the line on the QQ plot, with a small tails on the edge, however the majority of residuals lie upon the line allowing me to conclude that they are normally distributed. Lastly, residuals are homoscedastic, as they are randomly distributed in both the Scale-location and Residuals vs Fitted graphs, and outliers do not strongly affect the model, as seen by the lack of major outliers on the residuals vs. leverage plot.

Running the Test

```
# running a correlation test
# formula: (data frame$response, data frame$predictor)
cor.test(kelp$kelp_elong, kelp$temp_c,
         method = "pearson") # type of test: pearson (normal residuals from QQ)
```

Pearson's product-moment correlation

```
data: kelp$kelp_elong and kelp$temp_c
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
```

```
95 percent confidence interval:
-0.8359665 -0.4460157
sample estimates:
cor
-0.6877489
```

d. Results communication

To evaluate the strength of the relationship between temperature and giant kelp frond elongation rate, I used a Pearson's product-moment correlation because the data meets the assumptions of normality (as shown by the QQ residuals plot), we are observing a linear relationship between two continuous variables, and there are independent observations. We found a moderate negative relationship between ocean temperature and Giant Kelp Frond elongation rate (Pearson's $r = -0.688$, 95% CI $[-0.8359665, -0.4460157]$, $t(30) = -5.189$, $p < 0.001$, $\alpha = 0.05$).

e. Test implications

Our tests suggest that as ocean temperature increases, the growth rate of Giant Kelp decreases. In regards to Giant Kelp health and prosperity, it is important to continue to monitor these postulation in order to determine if temperature is affecting anything else, other than their growth rate. Such research is extremely important due to rising global temperatures in our oceans, and we must work towards accommodating these species who are negatively affected by the rising ocean temperatures.

f. Double check your own work.

```
# running the other correlation test
# formula: (data frame$response, data frame$predictor)
cor.test(kelp$kelp_elong, kelp$temp_c,
         method = "spearman") # type of test: spearman rank correlation
```

```
Warning in cor.test.default(kelp$kelp_elong, kelp$temp_c, method = "spearman"):
Cannot compute exact p-value with ties
```

Spearman's rank correlation rho

```
data: kelp$kelp_elong and kelp$temp_c
S = 9216.1, p-value = 1.29e-05
```

```
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
-0.6891684
```

The two tests would have led me to the same conclusion, which was to reject the null hypothesis and infer that there is a moderate negative correlation between ocean temperature and Giant Kelp Frond elongation rate (Spearman $\rho = -0.689$, $S = 9216.1$, $p < 0.001$, $\alpha = 0.05$). Despite gathering the same conclusion and results, the two tests are different in what they are comparing. The Pearson's product-moment correlation compares continuous variables and is the parametric version of a correlation test and a Spearman's rank correlation compares ranks and is the non-parametric version of a correlation test.

Problem 2. Personal data

a. Updating your visualizations

Weekend vs. Weekday Step Count Plot

```
# cleaning data first
my_data_clean <- my_data |>
  # cleaning column names
  clean_names() |>
  select(weekday_or_weekend, #selecting 1st column of interest
         step_count) # selecting next column of interest

# base layer: ggplot
ggplot(data = my_data_clean,
       mapping = aes(x = weekday_or_weekend, # x-axis = category (weekend/day)
                     y = step_count, # y-axis is step count
                     color = weekday_or_weekend)) + # assign category colors

# first layer: boxplot
geom_boxplot() +

# second layer: jitter plot
geom_jitter(height = 0, # making sure points don't move along y-axis
            width = 0.2) + # narrowing width of jitter

# labeling x-axis, y-axis, and adding a title
labs(x = "Weekday or Weekend",
     y = "Daily Step Count",
     title = "Number of Steps Taken differs on Weekday vs. Weekend") +

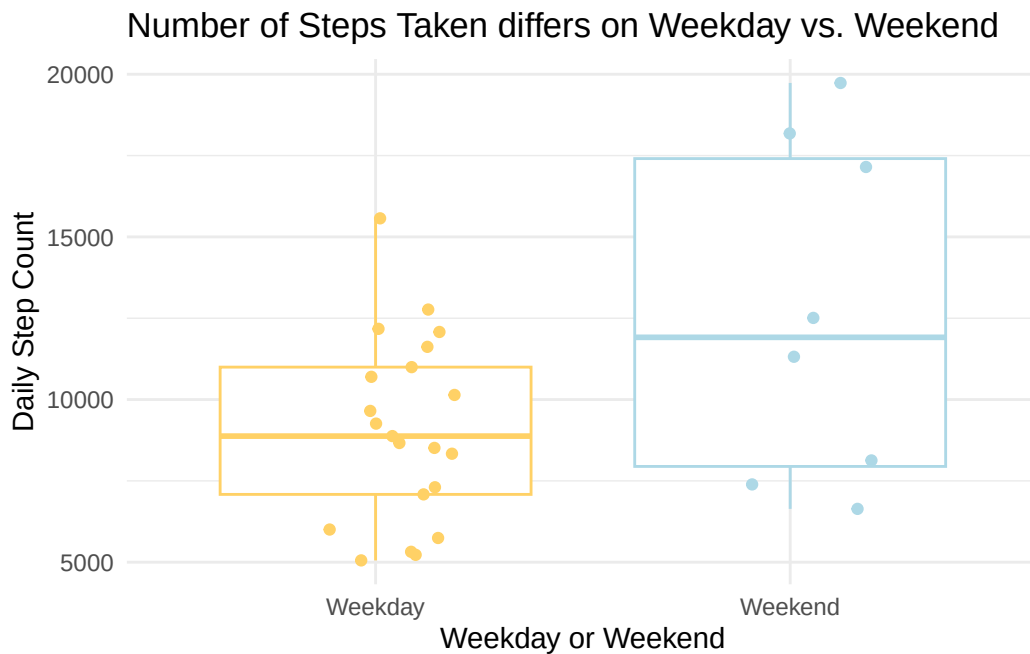
# changing the theme to differ from default
```

```

theme_minimal() +
# assigning colors based on category
scale_color_manual(values = c("Weekday" = "#FFD166",
                              "Weekend" = "lightblue")) +

# removing the legend
theme(legend.position = "none")

```



Daily High Temperature vs. Step Count

```

# cleaning data first
my_data_clean2 <- my_data |>
  clean_names() |> # cleaning column names
  select(step_count, #selecting 1st column of interest
         high_temperature_o_f) # selecting next column of interest

# fitting the linear model to generate the ribbon + linear line
my_data_model <- lm(
  step_count ~ high_temperature_o_f, # formula: step count as a function of temp
  data = my_data_clean2) # data from the kelp data set

# generating predictor values for the linear model + ribbon

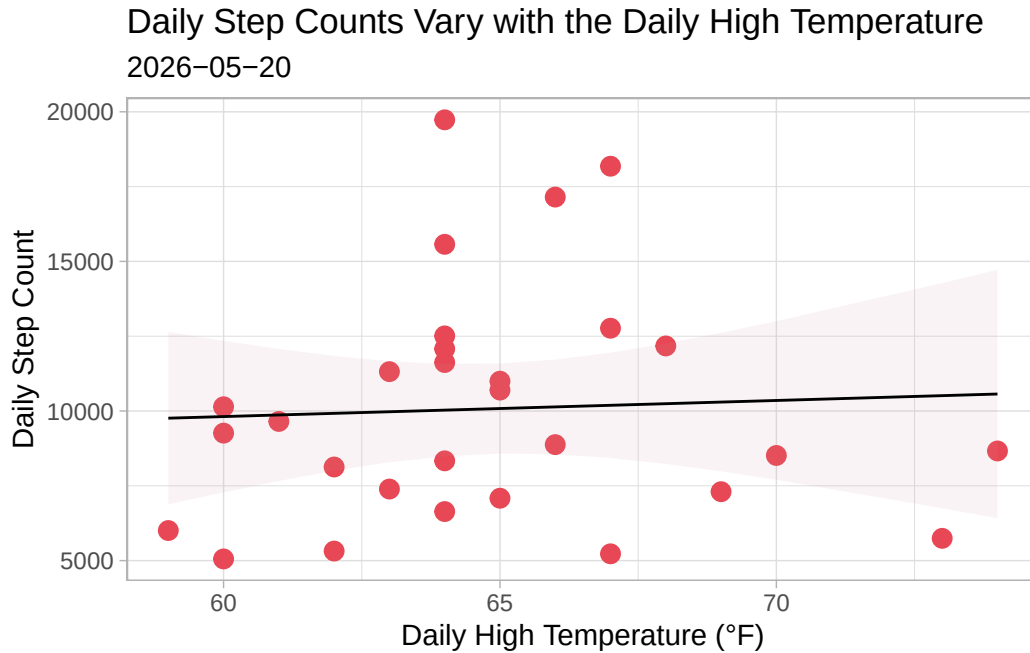
```

```

my_data_preds <- ggpredict(
  my_data_model,      # model object
  terms = "high_temperature_o_f") # predictor (in quotation marks)

# generating the plot
ggplot(data = my_data_clean2, # selecting the data frame
  mapping = aes(x = high_temperature_o_f, # x-axis is step count
    y = step_count, # y-axis is the day's high temperature
  )) +
  # first layer: geom_point
  geom_point(color = "#E84855", # adding colored visual data points
    size = 3) + # increaisng size of point
  # second layer: ribbon representing confidence interval
  # using predictions data frame
  geom_ribbon(data = my_data_preds, # data from predicted data set
    aes(x = x, # x = x in the data set
      y = predicted, # y = y in the data set
      ymin = conf.low, # min y point on ribbon set by CI
      ymax = conf.high), # max y point on ribbon set by CI
    alpha = 0.1, # opaqueness of the ribbon
    fill = "pink3") + # changing ribbon color
  # third layer: line representing model predictions
  # using predictions data frame
  geom_line(data = my_data_preds, # data from predicted data set
    aes(x = x, # x = x in the data set
      y = predicted)) + # y = y in the data set
  # relabel x- and y-axes, legend title (still want title?)
  labs(x = "Daily High Temperature (°F)",
    y = "Daily Step Count",
    title = "Daily Step Counts Vary with the Daily High Temperature",
    subtitle = "2026-05-20") +
  # custom themes
  theme_light()

```

b. Captions

Caption for “Number of Steps Taken differs on Weekday vs. Weekend”

Figure 1. Number of steps taken differs on Weekday versus weekend. Small circular points represent the total number of steps recorded on given weekday (yellow points) or weekend (blue points) from 2026-04-21 to 2026-05-20. Box plots represent median number of steps taken. The box itself represents the interquartile range, the horizontal line in the center represents the median number of steps taken on each day type, and the whiskers extend to the 1.5 x the IQR, and the whiskers extend to the 1.5 x the IQR, while anything outside that range is an outlier. Data collected by Mila Matich from 2026-04-21 to 2026-05-20.

Caption for “Daily Step Counts Vary with the Daily High Temperature”

Figure 2. Daily step count slightly varies (or weekly correlated?) with the Daily High Temperature. Red dots represents the number of steps taken under a certain daily high temperature. Pink ribbon represents the 95% confidence interval, and the black line represents the line of best fit. Data collected by Mila Matich from 2026-04-21 to 2026-05-20

Problem 3. Affective visualization

- a. Describe in words what an affective visualization could look like for your personal data (3-5 sentences).
- b. Create a sketch (on paper) of your idea.
- c. Make a draft of your visualization.
- d. Write an artist statement.
- e. Prep your materials to share in class.

Problem 4. Statistical critique

- a. Revisit and summarize
- b. Visual clarity
- c. Aesthetic clarity
- d. Recommendations